Sonar Net

OSINT-Based Username Reconnaissance using Kali Linux

Problem Statement:

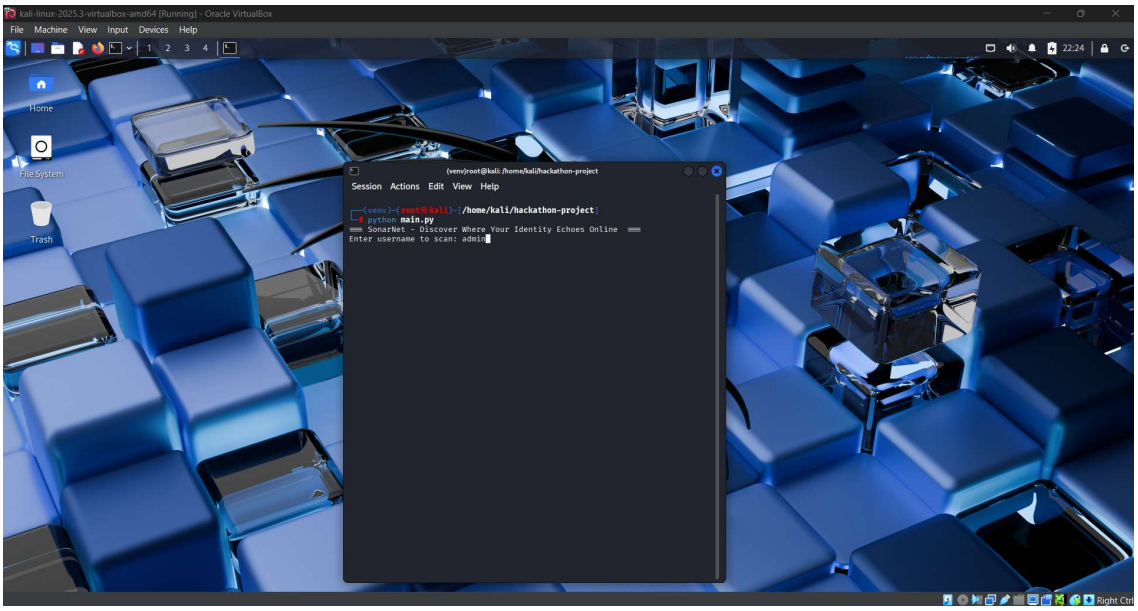
Sonar Net is a cybersecurity awareness and digital footprint analysis tool designed to identify and evaluate the public exposure of online identities. The system leverages Open-Source Intelligence (OSINT) techniques and security utilities available in Kali Linux to scan the internet for occurrences of a given username across multiple platforms and public databases. By automating reconnaissance using OSINT tools, SonarNet detects publicly accessible accounts, services, and digital traces associated with a specific identifier.

The platform aggregates this data and analyzes the breadth of online presence to determine how easily an individual's identity can be traced across the web. Based on the number and type of detected platforms, the system computes a digital exposure risk score, which represents the potential vulnerability of a user to privacy breaches, profiling, or social engineering attacks. The results are then compiled into a structured report that highlights discovered platforms, potential risks, and recommendations for improving digital hygiene.

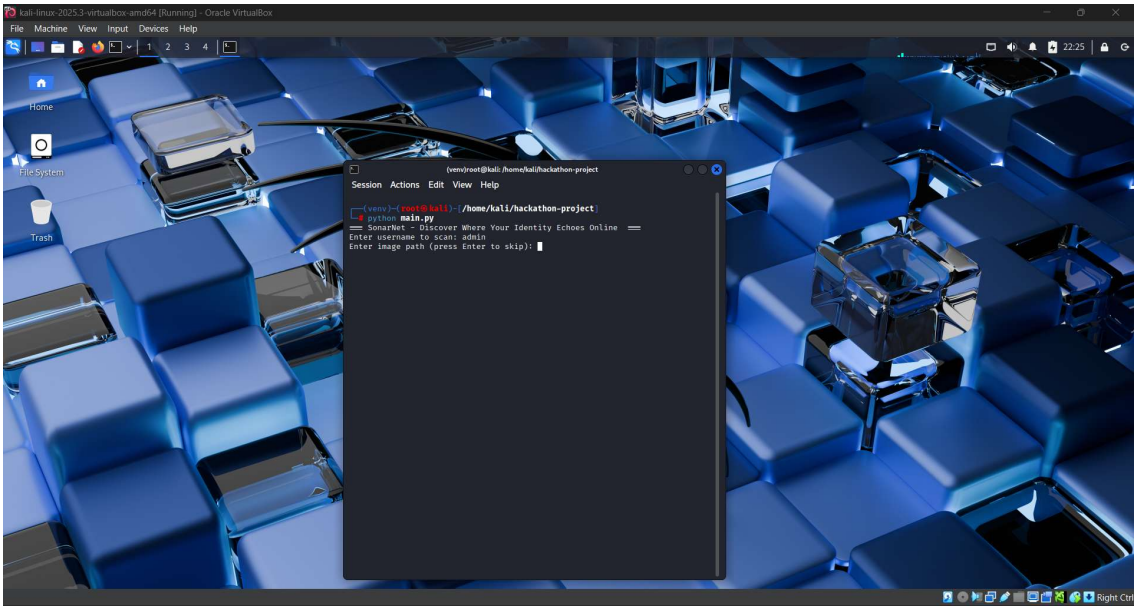
SonarNet serves as both a cybersecurity awareness tool and an educational resource, demonstrating how publicly available information can be systematically gathered and analyzed using OSINT methodologies. By visualizing the extent of a user's online footprint, the system encourages safer digital practices such as reducing username reuse, limiting publicly accessible data, and strengthening privacy protections. This project illustrates how readily accessible security tools can be used to better understand the risks associated with modern digital identities.

Implementation: -

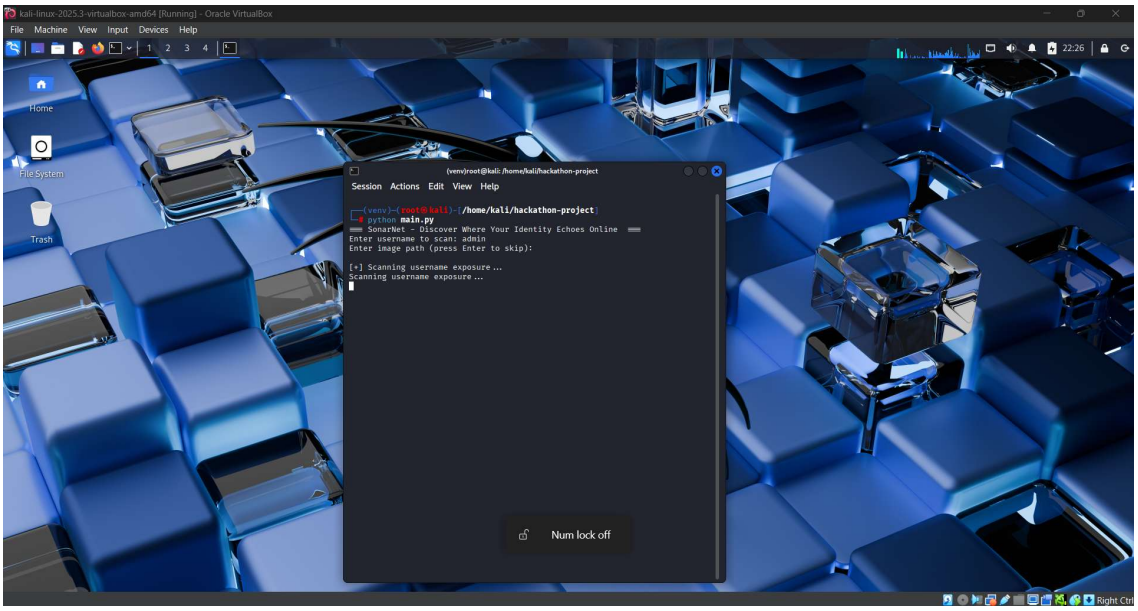
Step 1: -



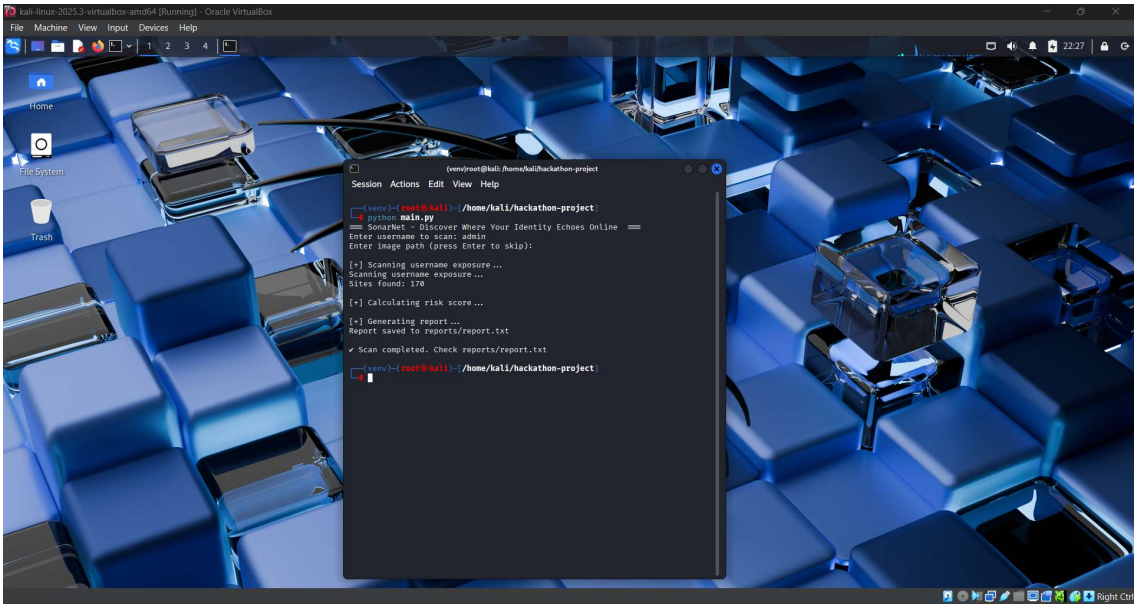
Step 2: -



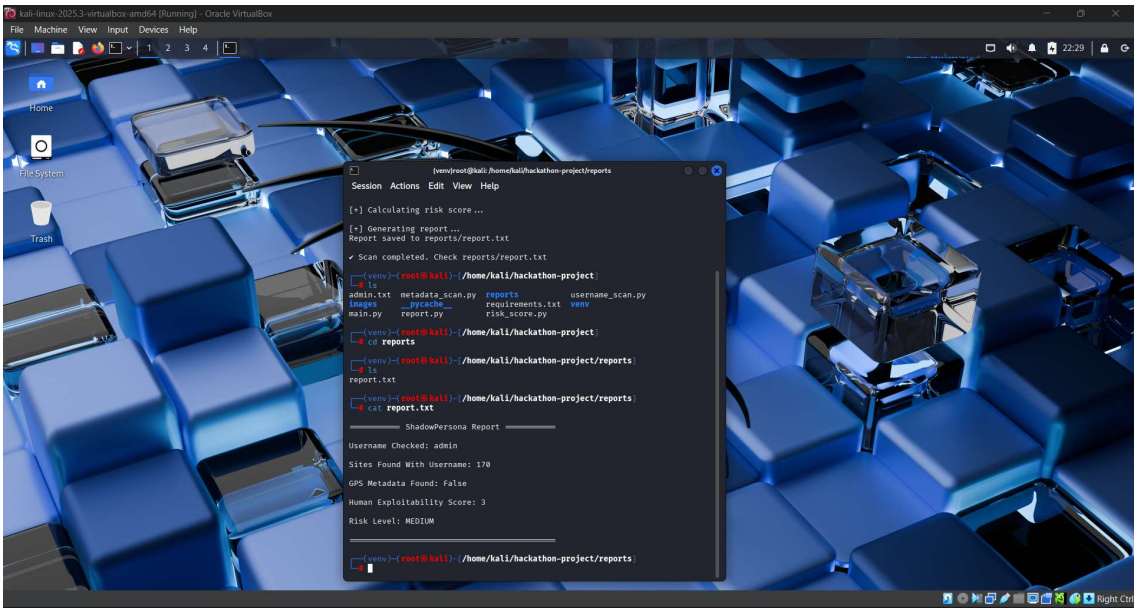
Step 3: -



Step 4: -



Step 5 (Output): -


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Home
File System
Trash
(venv)root@kali: /home/kali/hackathon-project/reports
Session Actions Edit View Help
[+] Calculating risk score ...
[+] Generating report ...
Report saved to reports/report.txt
✔ Scan completed. Check reports/report.txt
┌──(venv)─(root㉿kali)-[/home/kali/hackathon-project]
└─# ls
admin.txt metadata_scan.py reports username_scan.py
images __pycache__ requirements.txt venv
main.py report.py risk_score.py
┌──(venv)─(root㉿kali)-[/home/kali/hackathon-project]
└─# cd reports
┌──(venv)─(root㉿kali)-[/home/kali/hackathon-project/reports]
└─# ls
report.txt
┌──(venv)─(root㉿kali)-[/home/kali/hackathon-project/reports]
└─# cat report.txt
ShadowPersona Report
Username Checked: admin
Sites Found With Username: 170
GPS Metadata Found: False
Human Exploitability Score: 3
Risk Level: MEDIUM
┌──(venv)─(root㉿kali)-[/home/kali/hackathon-project/reports]
└─#
22:29
Right Ctrl